TT TIMETRADE PLATFORM

BCA Sem V • Capstone Project

FULL-STACK ACADEMIC PEER-TO-PEER PLATFORM

# TimeTrade: Peer-to-Peer Academic Assistance & Slot Reservation

Reciprocal time-banking ecosystem powered by MERN Stack (MongoDB, Express, React 18, Node.js) and Tailwind CSS.


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ACADEMIC GUIDE

CS & IT Evaluation Committee


ACADEMIC YEAR

2025 – 2026

01. EXECUTIVE SUMMARY

# Eliminating Campus Peer-Learning Friction

### The Reciprocal Time-Bank Core

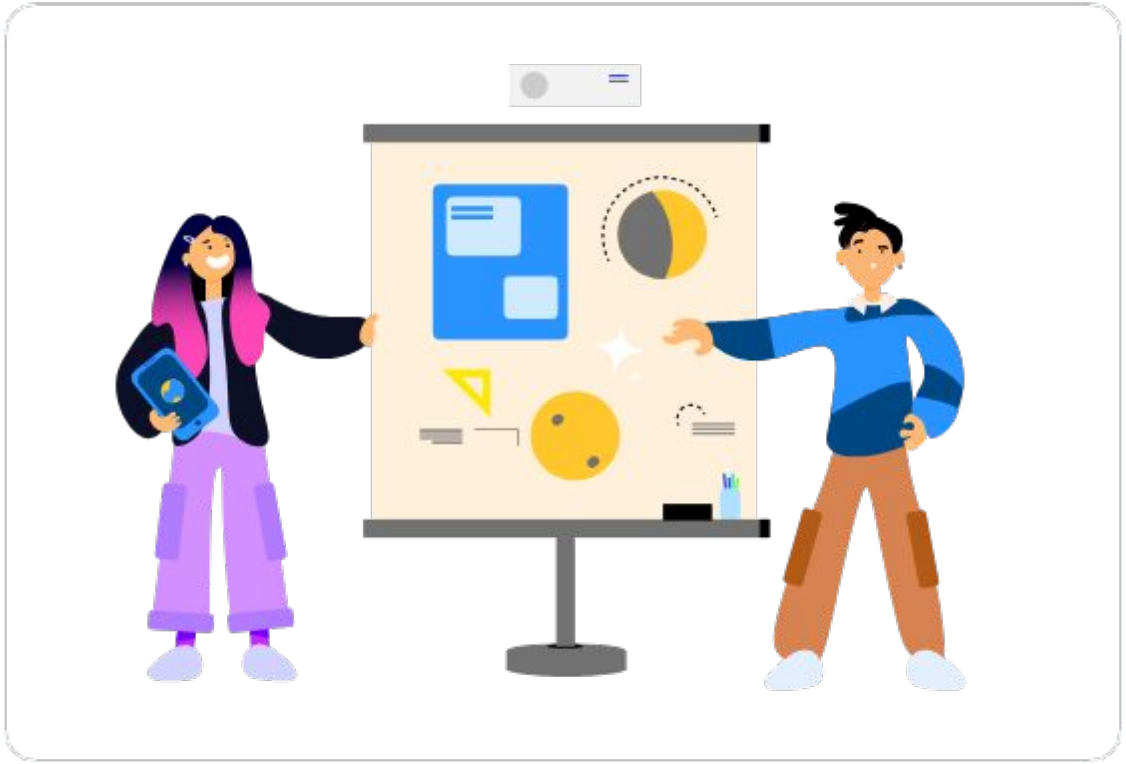
1 hour of React debugging provided earns 1 hour of mock pre-defense preparation received, removing monetary transaction barriers.

**Decoupled SPA**

React 18 + Vite client isolated from Node/Express API backend.

**State Machine**

Deterministic Open → Booked → Completed session lifecycles.



02. PROBLEM STATEMENT

# Academic Collaboration Bottlenecks

### Fragmented Channels

Academic help queries lost in unindexed instant messaging apps.

### Assistance Burnout

High performers experience fatigue without reciprocal academic support.

### Revision Bottlenecks

Lack of structured, peer-curated lecture summaries before exams.

### Defense Anxiety

Suboptimal capstone defense performance due to missing mock practice.

03. MARKETPLACE TAXONOMY

# Supported Exchange Categories

| Category | Typical Scope | Target Student Need | Standard Slot |
|---|---|---|---|
| **Coding** | React, Node.js, Python, DSA | Bug triage & code walkthroughs | **30 – 60 Mins** |
| **Notes** | Lecture summaries, PYQs, sheets | Midterm & semester exam revision | **30 – 45 Mins** |
| **Presentation** | Defense rehearsal, slide feedback | Final presentation & Q&A readiness | **45 – 60 Mins** |
| **Academic Support** | Discrete Math, Statistics, Automata | Complex proofs & theory logic | **45 – 90 Mins** |
| **Other** | Git, Linux CLI, Docker setup | DevOps toolchain pairing | **30 – 45 Mins** |

04. ENGINEERING STACK

# Production Technology Choices

Frontend UI

## React 18.3 & Vite

Declarative UI with instant HMR bundling and custom Hook state management.

Styling Engine

## Tailwind CSS 3.4

Utility-first responsive framework with zero runtime style sheet overhead.

Backend API

## Node 24 & Express

Asynchronous non-blocking REST gateway handling stateless requests.

Database

## MongoDB Mongoose

Document storage with compound index constraints and populated refs.

Routing & Guards

## React Router 6.28

Protected client routes verifying authentication token persistence.

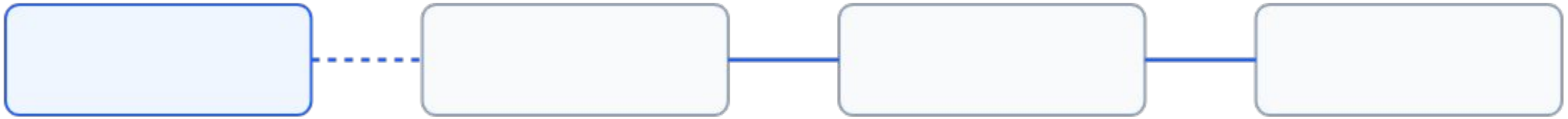
Network Client

## Axios Pipeline

Centralized HTTP instance with automatic token injection and 401 handling.

05. SYSTEM ARCHITECTURE

# Layered Request Flow & Gateway



**1. Client Presentation**
React SPA + Axios Bearer Header Injection

**2. Security Gateway**
Helmet, Rate Limit & Express Validator

**3. Business Service**
Domain Controllers & Booking State Engine

**4. Persistence Layer**
MongoDB Compound Indexes & Schemas

06. DATA ARCHITECTURE

# MongoDB Entity Schema Design

### User Entity `User.js`

**email:** Unique index, lowercase

**password:** Bcrypt hashed (select: false)

**department:** Academic major string

**avatarUrl:** Dynamic SVG fallback

### Session Entity `Session.js`

**creator:** User ObjectId reference

**title / description:** Full text index

**category:** Enum [Notes, Coding, Pres...]

**status:** Compound indexed availability

### Booking Entity `Booking.js`

**session / requester:** Compound unique

**message:** Student learning objectives

**status:** Pending | Accepted | Rejected

**timestamps:** Automated audit dates

07. CORE LOGIC

# Two-Sided Request State Machine

```
// bookingService.js - Host Acceptance Flow
if (newStatus === 'Accepted') {
  // 1. Mark target session as 'Booked'
  await Session.findByIdAndUpdate(session, { status: 'Booked' });
  // 2. Reject all other pending requests
  await Booking.updateMany(
    { session, _id: { $ne: bookingId }, status: 'Pending' },
    { status: 'Rejected' }
  );
}
```

### Atomic Host Acceptance

Accepting one inquiry instantly reserves the slot and clears competing queue requests.

### Automatic Availability Reversal

Cancelling an accepted booking reverts the parent session status back to 'Open'.

08. SECURITY ENGINEERING

# OWASP Aligned Hardening Measures

### Bcrypt Salt Hashing

10 key-derivation rounds protecting user passwords against rainbow tables.

### Helmet Header Policy

CSP headers, anti-clickjacking frameguards, and HTTP HSTS enforcement.

### Rate Limit Throttling

IP-based limiting (30 req / 15 min) shielding authentication routes.

### Input Sanitization

Express-validator schema checks neutralizing NoSQL injection vectors.

### Origin Whitelisting

Strict CORS policies restricting cross-origin AJAX access to trusted domains.

### Error Isolation

Global error handling concealing backend stack traces from API payloads.

09. QUALITY ASSURANCE

# API Integration Test Matrix

| Test ID | Target Endpoint | Scenario & Conditions | Expected | Status |
|---|---|---|---|---|
| TC-AUTH-01 | POST /api/auth/register | Valid institutional registration payload | **201 Created** | PASS |
| TC-AUTH-02 | POST /api/auth/register | Duplicate email registration attempt | **409 Conflict** | PASS |
| TC-BOOK-01 | POST /api/bookings | Student requesting own session offering | **400 Bad Req** | PASS |
| TC-BOOK-02 | PATCH /api/bookings/:id | Host accepts slot; auto-reject competing | **200 OK** | PASS |
| TC-BOOK-03 | PATCH /api/bookings/:id/cancel | Accepted booking cancelled by requester | **200 OK** | PASS |

10. PRODUCT EVOLUTION

# Future Enhancement Roadmap

PHASE 01

### WebSocket Chat

Socket.io integration for instant pre-booking logistics.

PHASE 02

### WebRTC Video

In-browser video study rooms with screen sharing.

PHASE 03

### Calendar Sync

Automated export of sessions to Google Calendar & .ics.

PHASE 04

### Karma Ledger

Digital credit balances with peer ratings and endorsements.



# Thank You!

TimeTrade Project Defense & Evaluation Ready

REPOSITORY
**GitHub / timetrade-platform**

ENVIRONMENT
**Production MERN Ready**